

energy is related to wavelength (or colour), this means that the optical properties of the particle can be finely tuned depending on its size. Thus, particles can be made to emit or absorb specific wavelengths (colours) of light, merely by controlling their size.

Quantum dots have found applications in composites, solar cells, fluorescent biological labels etc. Recent advances in chemistry have resulted in the preparation of monolayer-protected, high-quality, monodispersed, crystalline quantum dots as small as 2nm in dia, which can be conveniently treated and processed as a typical chemical reagent.

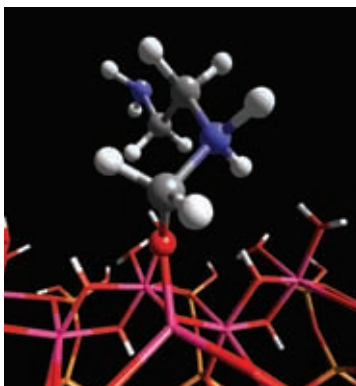
Nanocomposites

A nanocomposite is a matrix to which nanoparticles have been added to improve a particular property of the material. It is a multiphase solid material where one of the phases has one, two or three dimensions of less than 100 nanos (nm), or structures having nano-scale repeat distances between the different phases that make up the material.

In the broadest sense nanocomposite include porous media, colloids, gels and copolymers. Understandably the mechanical, electrical, thermal, optical, electrochemical, catalytic properties of the nanocomposite will differ markedly from that of the component materials. Size limits for these effects have been proposed as

- ▶ Less than 5 nm for catalytic activity,
- ▶ Less than 20 nm for making a hard magnetic material soft
- ▶ Less than 50 nm for refractive index changes, and
- ▶ Less than 100 nm for achieving superparamagnetism, mechanical strengthening or restricting matrix dislocation movement.

In mechanical terms, nanocomposites differ from conventional composite materials due to the exceptionally high surface to volume ratio of the reinforcing phase and/or its exceptionally high aspect ratio. The area of the interface between the matrix and reinforcement phase(s) is typically an order of magnitude greater than for conventional composite materials. The matrix material properties are significantly affected in the vicinity of the reinforcement. The large amount of



An artist's rendition of nanocomposite